Year 11 Set 4 · Foundation · Lesson 6

Probability

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Today I will learn to...

Goals

1. Use the fact that probabilities add up to 1

2. Work out expected outcomes and relative frequency

Do now: retrieval practice

4 mins

1. Hypotenuse of a triangle with sides 6 and 8

   10

2. Work out 0.3 + 0.45

   0.75

3. Work out 1 – 0.35

   0.65

4. Find 2/5 of 30

   12

Objective 1

Use the fact that probabilities add up to 1

Key idea

Learn

!

All the probabilities for one event add up to 1. P(not A) = 1 – P(A).

Objective 1: Use the fact that probabilities add up to 1

I do: example 1

I do

P(red) = 0.3. Find P(not red).

1. $1 - 0.3$
2. $= 0.7$

I do: example 2

I do

Red 0.2, blue 0.35, green x, yellow 0.15. Find x.

1. $0.2 + 0.35 + 0.15 = 0.7$

2. $x = 1 - 0.7 = 0.3$

I do: example 3

I do

A bag has 3 red, 5 blue and 2 green counters. Find P(blue).

1. 10 counters in total

2. P(blue) = 5/10 = 1/2

Mini whiteboards

We do

1. P(win) = 0.62. P(lose)?

Answer: 0.38

2. A: 0.4, B: 0.25, C: ?

Answer: 0.35

3. P(rain) = 0.7. P(no rain)?

Answer: 0.3

4. A: 0.15, B: 0.5, C: 0.2, D: ?

Answer: 0.15

You do: choose your challenge

You do

Bronze

A team can only win or lose. P(lose) = 0.45. Find P(win).

Silver

A spinner is red, blue or green. P(red) = 0.25, P(blue) = 0.4. Find P(green).

Gold

A bag has red, blue and white counters. P(red) = 0.3, P(blue) = 0.5. There are 8 white counters. How many counters are in the bag?

Platinum

P(red) = 2x, P(blue) = 3x, P(green) = 0.25. These are the only colours. Find x.

You do: answers

Check

Bronze	0.55
Silver	0.35
Gold	P(white) = 0.2, 8 ÷ 0.2 = 40
Platinum	5x = 0.75, x = 0.15

If you finish early

Extra

Bronze

P(goal) = 0.35. Find P(no goal).

0.65

Silver

P(red) = 0.1, P(blue) = 0.45, P(green) = 0.3. Yellow is the only other colour. Find P(yellow).

0.15

Gold

A spinner has P(1) = 0.2, P(2) = 0.3 and P(3) = P(4). Find P(4).

0.25

Platinum

A bag has only red and blue counters. P(red) = 3/8. There are 15 blue counters. How many red counters are there?

Blue = 5/8, so 1/8 = 3 counters. Red = 9

Exam question 1

2 marks

A dice is biased. P(1) = 0.1, P(2) = 0.25, P(3) = 0.2, P(4) = x, P(5) = x, P(6) = 0. Find x.

Space for working

Mark scheme: 0.55 + 2x = 1 → x = 0.225

Exam question 2

1 mark

A bag has 4 red and 6 yellow counters. One is taken at random. Find P(yellow).

Space for working

Mark scheme: 6/10 = 3/5

Objective 1 achieved

Use the fact that probabilities add up to 1



Objective 2

Work out expected outcomes and relative frequency

✓ Achieved: Use the fact that probabilities add up to 1

Key idea

Learn

!

Expected number = probability × number of trials.
Relative frequency = times it happened ÷ total trials.

Objective 2: Work out expected outcomes and relative frequency

I do: example 1

I do

P(rain) = 0.2. How many rainy days in 30?

1. 0.2×30
2. = 6 days

I do: example 2

I do

A fair coin is flipped 200 times. Expected heads?

1. 0.5×200
2. $= 100$

I do: example 3

I do

A dice is rolled 50 times and lands on 6 fifteen times. Is it fair?

1. Relative frequency = 15/50 = 0.3
2. Expected about 50 ÷ 6 ≈ 8 sixes
3. 15 is much more, so it may be biased

Mini whiteboards

We do

1. P = 0.15, 60 trials. Expected?

Answer: 9

2. Red 18 times out of 40. Relative frequency?

Answer: 0.45

3. Fair coin, 80 flips. Expected heads?

Answer: 40

4. 6 sixes in 50 rolls. Relative frequency?

Answer: 0.12

You do: choose your challenge

You do

Bronze

P = 0.3. How many times would you expect it in 50 trials?

Silver

A fair dice is rolled 120 times. How many 5s do you expect?

Gold

A spinner lands on red 35 times in 140 spins. Find the relative frequency, and estimate the number of reds in 400 spins.

Platinum

P(Tia is late) = 0.08 each day. School runs 5 days a week for 38 weeks. Estimate how many days she is late.

You do: answers

Check

Bronze

15

Silver

20

Gold

0.25, so about 100

Platinum

190 × 0.08 = 15.2, about 15 days

If you finish early

Extra

Bronze

P = 0.2. Expected number in 200 trials?

40

Silver

A fair dice is rolled 300 times. How many even numbers do you expect?

150

Gold

A drawing pin lands point up 27 times in 60 throws. Estimate how many times in 500 throws.

0.45 × 500 = 225

Platinum

A spinner should land on red with P = 0.3. In 250 spins it lands on red 40 times. Is it fair? Explain.

Expect 75 reds, only got 40, so it is probably biased

Exam question 1

2 marks

P(a biased coin lands heads) = 0.65. It is thrown 120 times. Estimate the number of heads.

Space for working

Mark scheme: 0.65 × 120 = 78

Exam question 2

1 mark

Lisa rolls a dice 300 times and gets a 4 sixty-two times. Estimate P(4).

Space for working

Mark scheme: 62/300 ≈ 0.21

Objective 2 achieved

Work out expected outcomes and relative frequency



Real life

Apply

A raffle sells 500 tickets and you buy 4. What is the probability you win first prize?

4/500 = 1/125

Extension: stretch yourself

Stretch

1. Two coins are flipped. Find P(two heads).

HH, HT, TH, TT → 1/4

2. Two dice are rolled. Find P(total = 7).

6 ways out of 36 = 1/6

Exit ticket: exam questions

Exit

1. P(snow) = 0.12. Find P(no snow). **(1 mark)**

2. A bag has 3 red and 7 green balls. One is taken at random. Find P(red). **(1 mark)**

3. P(a biased coin lands heads) = 0.6. It is thrown 250 times. Work out the expected number of heads. **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 0.88 [1]

Q2 3/10 [1]

Q3 150 [2]

Total: ____ / 4

Today I achieved

- ✔ **Use the fact that probabilities add up to 1**
- ✔ **Work out expected outcomes and relative frequency**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. A fair spinner is numbered 1 to 8. Find P(even number). [1]

2. P(A) = 0.15, P(B) = 0.32, P(D) = 0.2. A, B, C and D are the only outcomes. Find P(C). [2]

3. A bag has 5 red, 4 blue and 1 yellow counter. Find P(not blue). [2]

4. A dice is rolled 80 times and lands on 6 twenty-two times. Find the relative frequency of a 6. [1]

5. P(a train is late) = 0.15. Estimate how many of 240 trains will be late. [2]

Extra practice: Maths Genie → Edexcel question bank → Probability